POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6034A | Semester VI | Theory | 4 Credits

Solar Power Technologies

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6034A as Solar Power Technologies in Semester VI for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Guwahati’s Solar Energy Engineering and Technology course and polytechnic solar technology curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Solar designs, PV installations, thermal systems and structural mountings must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support solar radiation analysis, PV technology, solar thermal conversion, and system sizing. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Solar Power Technologies studies the principles, design and application of solar energy conversion systems. It develops the ability to analyze solar radiation, evaluate photovoltaic (PV) and solar thermal technologies, design stand-alone and grid-connected solar power systems, and coordinate renewable energy installations while respecting the technical and safety boundaries of electrical and solar engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□□ □□□□□□□□□□□; □□□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□□ □□□□□□□□□□.

Prerequisite foundation

Basic electrical engineering, semiconductor physics, heat transfer and energy conversion principles.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamentals of solar radiation, solar geometry and measurement techniques.
2. Analyze the operation, characteristics and fabrication of various solar photovoltaic (PV) cells and modules.
3. Explain the principles and applications of solar thermal conversion systems including collectors and power plants.
4. Design and size solar PV systems for residential and industrial applications including battery and grid integration.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain solar radiation characteristics and measurement techniques.	Understand
CO2	Analyze the performance and technology of solar photovoltaic cells and modules.	Apply
CO3	Explain the operation and applications of solar thermal collectors and systems.	Understand
CO4	Design and size stand-alone and grid-connected solar PV systems.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Solar Radiation and Measurement	Solar constant; Solar radiation: Extraterrestrial and Terrestrial; Solar geometry: Zenith, Azimuth, Tilt angles; Measurement: Pyranometer, Pyrheliometer; Estimation of solar radiation on tilted surfaces.	Approx. 14 hours	CO1	Module 1
2	Solar Photovoltaic (PV) Technology	Solar cell physics: P-N junction, I-V characteristics, Fill factor, Efficiency; PV module, panel and array; Cell types: Monocrystalline, Polycrystalline, Thin film; MPPT and Inverters.	Approx. 16 hours	CO2	Module 2
3	Solar Thermal Systems	Principles of solar thermal conversion; Flat plate collectors (FPC); Concentrating collectors: Parabolic trough, Dish, Tower; Solar water heating; Solar thermal power plants (CSP).	Approx. 14 hours	CO3	Module 3
4	PV System Design and Integration	Stand-alone PV system design: Load estimation, Battery sizing, Charge controller; Grid-connected systems: Net metering, Grid-tie inverters; Balance of System (BOS); Installation and safety.	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Solar Radiation and Measurement

Solar constant; Solar radiation: Extraterrestrial and Terrestrial; Solar geometry: Zenith, Azimuth, Tilt angles; Measurement: Pyranometer, Pyrheliometer; Estimation of solar radiation on tilted surfaces.

CO1 · Approx. 14 hours

Solar Photovoltaic (PV) Technology

Solar cell physics: P-N junction, I-V characteristics, Fill factor, Efficiency; PV module, panel and array; Cell types: Monocrystalline, Polycrystalline, Thin film; MPPT and Inverters.

CO2 · Approx. 16 hours

Solar Thermal Systems

Principles of solar thermal conversion; Flat plate collectors (FPC); Concentrating collectors: Parabolic trough, Dish, Tower; Solar water heating; Solar thermal power plants (CSP).

CO3 · Approx. 14 hours

PV System Design and Integration

Stand-alone PV system design: Load estimation, Battery sizing, Charge controller; Grid-connected systems: Net metering, Grid-tie inverters; Balance of System (BOS); Installation and safety.

CO4 · Approx. 16 hours

MODULE 1

Solar Radiation and Measurement

Solar constant; Solar radiation: Extraterrestrial and Terrestrial; Solar geometry: Zenith, Azimuth, Tilt angles; Measurement: Pyranometer, Pyrheliometer; Estimation of solar radiation on tilted surfaces.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

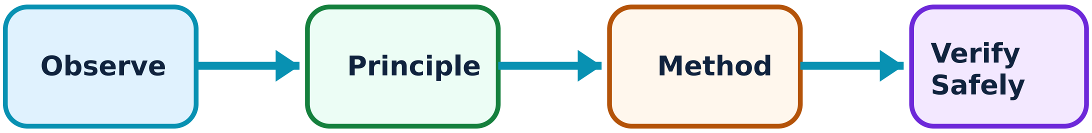
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Solar Radiation and Measurement: identify the system, state its principle, follow the correct method and verify the result safely.

1. Solar Radiation Fundamentals−

The Sun is a continuous source of energy. Solar radiation reaching the Earth's atmosphere is the solar constant (~1367 W/m²). Terrestrial radiation is affected by atmospheric absorption and scattering. Understanding solar angles (Zenith, Azimuth) is critical for optimal panel orientation.

Study and application method

Define 'Solar Constant'; explain the difference between 'Beam', 'Diffuse', and 'Global' radiation; list three solar geometry angles.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Solar Constant'; explain the difference between 'Beam', 'Diffuse', and 'Global' radiation; list three solar geometry angles.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Direct exposure to concentrated solar radiation can cause burns or eye damage. Do not use solar concentrators without authorised safety shielding and eye protection.

2. Measurement and Estimation−

Solar radiation is measured using a Pyranometer (for global radiation) and a Pyrheliometer (for beam radiation). Estimation models help in calculating the available solar energy on a tilted surface at a specific location and time.

Study and application method

Sketch the working principle of a 'Pyranometer'; explain why solar panels are usually tilted towards the South in the Northern Hemisphere.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

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Common mistake / precaution: Measurement instruments require precise calibration and leveling. Do not use solar radiation data for system sizing without authorised verification of the measurement site and instrument accuracy.

Energy-use calculator

Appliance power (W)

100

Hours used

5

Tariff per unit

6

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Solar Photovoltaic (PV) Technology

Solar cell physics: P-N junction, I-V characteristics, Fill factor, Efficiency; PV module, panel and array; Cell types: Monocrystalline, Polycrystalline, Thin film; MPPT and Inverters.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

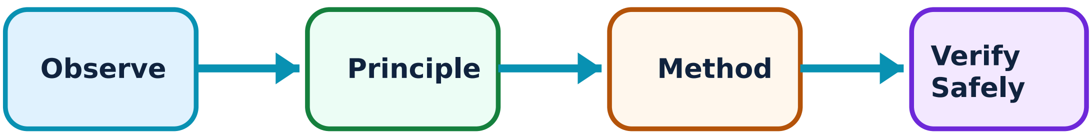
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Solar Photovoltaic (PV) Technology: identify the system, state its principle, follow the correct method and verify the result safely.

1. Solar Cell Operation−

A solar cell is a P-N junction that converts photons into electron-hole pairs. The I-V characteristic curve defines its performance, with key parameters being Short Circuit Current (Isc), Open

Circuit Voltage (Voc), and Fill Factor (FF). Efficiency is the ratio of electrical power output to incident solar power.

Study and application method

Sketch the 'I-V Characteristic Curve' of a solar cell and label Isc, Voc, and Pmax; define 'Fill Factor'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the 'I-V Characteristic Curve' of a solar cell and label Isc, Voc, and Pmax; define 'Fill Factor'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Solar cells are fragile and sensitive to shading. Do not handle cells without authorised protection or operate shaded modules without bypass diodes.

2. PV Modules and Electronics−

Cells are connected in series/parallel to form modules and arrays. Maximum Power Point Tracking (MPPT) is used to extract the maximum possible power under varying conditions. Inverters convert the DC output of the PV array into AC for use or grid injection.

Study and application method

Differentiate between 'Monocrystalline' and 'Polycrystalline' solar cells; explain the role of an 'Inverter' in a PV system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Monocrystalline' and 'Polycrystalline' solar cells; explain the role of an 'Inverter' in a PV system.

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Common mistake / precaution: PV modules generate DC voltage as soon as they are exposed to light. Do not work on PV wiring without authorised isolation and high-voltage DC safety precautions.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Solar Thermal Systems

Principles of solar thermal conversion; Flat plate collectors (FPC); Concentrating collectors: Parabolic trough, Dish, Tower; Solar water heating; Solar thermal power plants (CSP).

Approx. 14 hours **CO3** **Understand · Apply**

Learning goals

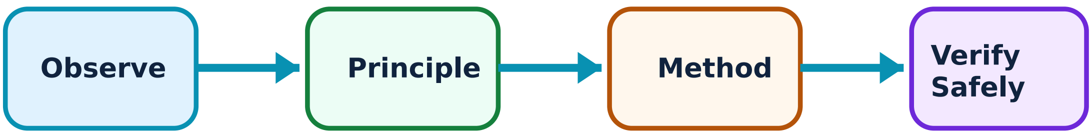
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Solar Thermal Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Solar Collectors−

Solar thermal systems collect heat from the sun. Flat plate collectors are used for low-temperature applications like water heating. Concentrating collectors (CSP) focus sunlight onto a receiver to achieve high temperatures for power generation.

Study and application method

Compare 'Flat Plate' and 'Concentrating' collectors conceptually; describe the working of a 'Parabolic Trough' collector.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Flat Plate' and 'Concentrating' collectors conceptually; describe the working of a 'Parabolic Trough' collector.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Solar thermal fluids can reach extremely high temperatures and pressures. Do not perform maintenance on thermal loops without authorised pressure relief and thermal protection.

2. Solar Thermal Applications−

Common applications include solar water heaters, solar cookers, and solar dryers. Large-scale solar thermal power plants use steam turbines to generate electricity, often with thermal energy storage (e.g., molten salt) for nighttime operation.

Study and application method

Sketch the block diagram of a 'Solar Water Heating System'; explain the concept of 'Thermal Energy Storage' in CSP plants.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of a 'Solar Water Heating System'; explain the concept of 'Thermal Energy Storage' in CSP plants.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Reflective surfaces in concentrating systems can cause accidental fires or blinding glares. All CSP sites must follow authorised optical safety and fire prevention protocols.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

PV System Design and Integration

Stand-alone PV system design: Load estimation, Battery sizing, Charge controller; Grid-connected systems: Net metering, Grid-tie inverters; Balance of System (BOS); Installation and safety.

Approx. 16 hours | **CO4** | **Understand · Apply**

Learning goals

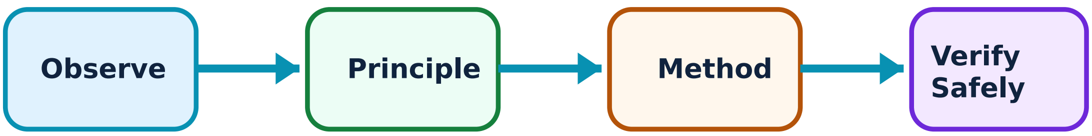
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for PV System Design and Integration: identify the system, state its principle, follow the correct method and verify the result safely.

1. Stand-alone System Design−

Off-grid systems require batteries to store energy for use when the sun is not shining. Sizing involves estimating the daily energy demand, determining the required PV array capacity, and selecting a battery bank and charge controller to prevent overcharging.

Study and application method

Calculate the required 'Battery Capacity' (Ah) for a given load and autonomy period; explain the role of a 'Charge Controller'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the required 'Battery Capacity' (Ah) for a given load and autonomy period; explain the role of a 'Charge Controller'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Lead-acid and Lithium batteries require proper ventilation and protection. Do not install battery banks in enclosed spaces without authorised gas venting and fuse protection.

2. Grid Integration and Safety−

Grid-connected systems use net metering to export excess power to the utility. The 'Balance of System' (BOS) includes mounting structures, cables, and protection devices. Safety is paramount during installation, requiring proper grounding and lightning protection.

Study and application method

Explain the concept of 'Net Metering'; list the main 'BOS' components of a grid-connected PV system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the concept of 'Net Metering'; list the main 'BOS' components of a grid-connected PV system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Grid-tied systems must automatically disconnect during a power outage (anti-islanding). Do not connect to the grid without authorised utility approval and certified grid-tie equipment.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Solar Radiation and Measurement

Use the concept flow to revise observation, principle, method and verification.

Module 2: Solar Photovoltaic (PV) Technology

Use the concept flow to revise observation, principle, method and verification.

Module 3: Solar Thermal Systems

Use the concept flow to revise observation, principle, method and verification.

Module 4: PV System Design and Integration

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Monocrystalline and Polycrystalline modules in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the path of the sun across the sky and label the Zenith and Azimuth angles.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of a Stand-alone PV system showing the PV array, charge controller, battery, and inverter.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the annual energy yield conceptually for a 1kW PV system with a given solar insolation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the solar panels on a nearby building and note their orientation and type if possible.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Solar Radiation and Measurement

Explain Solar Radiation Fundamentals and state one correct method, application or precaution.−

The Sun is a continuous source of energy. Solar radiation reaching the Earth's atmosphere is the solar constant (~1367 W/m²). Terrestrial radiation is affected by atmospheric absorption and scattering. Understanding solar angles (Zenith, Azimuth) is critical for optimal panel orientation.

Answer framework: Define 'Solar Constant'; explain the difference between 'Beam', 'Diffuse', and 'Global' radiation; list three solar geometry angles.

Important point: Direct exposure to concentrated solar radiation can cause burns or eye damage. Do not use solar concentrators without authorised safety shielding and eye protection.

Explain Measurement and Estimation and state one correct method, application or precaution.−

Solar radiation is measured using a Pyranometer (for global radiation) and a Pyrheliometer (for beam radiation). Estimation models help in calculating the available solar energy on a tilted surface at a specific location and time.

Answer framework: Sketch the working principle of a 'Pyranometer'; explain why solar panels are usually tilted towards the South in the Northern Hemisphere.

Important point: Measurement instruments require precise calibration and leveling. Do not use solar radiation data for system sizing without authorised verification of the measurement site and instrument accuracy.

Module 2 — Solar Photovoltaic (PV) Technology

Explain Solar Cell Operation and state one correct method, application or precaution. −

A solar cell is a P-N junction that converts photons into electron-hole pairs. The I-V characteristic curve defines its performance, with key parameters being Short Circuit Current (Isc), Open Circuit Voltage (Voc), and Fill Factor (FF). Efficiency is the ratio of electrical power output to incident solar power.

Answer framework: Sketch the 'I-V Characteristic Curve' of a solar cell and label Isc, Voc, and Pmax; define 'Fill Factor'.

Important point: Solar cells are fragile and sensitive to shading. Do not handle cells without authorised protection or operate shaded modules without bypass diodes.

Explain PV Modules and Electronics and state one correct method, application or precaution.−

Cells are connected in series/parallel to form modules and arrays. Maximum Power Point Tracking (MPPT) is used to extract the maximum possible power under varying conditions. Inverters convert the DC output of the PV array into AC for use or grid injection.

Answer framework: Differentiate between 'Monocrystalline' and 'Polycrystalline' solar cells; explain the role of an 'Inverter' in a PV system.

Important point: PV modules generate DC voltage as soon as they are exposed to light. Do not work on PV wiring without authorised isolation and high-voltage DC safety precautions.

Module 3 — Solar Thermal Systems

Explain Solar Collectors and state one correct method, application or precaution.−

Solar thermal systems collect heat from the sun. Flat plate collectors are used for low-temperature applications like water heating. Concentrating collectors (CSP) focus sunlight onto a receiver to achieve high temperatures for power generation.

Answer framework: Compare 'Flat Plate' and 'Concentrating' collectors conceptually; describe the working of a 'Parabolic Trough' collector.

Important point: Solar thermal fluids can reach extremely high temperatures and pressures. Do not perform maintenance on thermal loops without authorised pressure relief and thermal protection.

Explain Solar Thermal Applications and state one correct method, application or precaution.−

Common applications include solar water heaters, solar cookers, and solar dryers. Large-scale solar thermal power plants use steam turbines to generate electricity, often with thermal energy storage (e.g., molten salt) for nighttime operation.

Answer framework: Sketch the block diagram of a 'Solar Water Heating System'; explain the concept of 'Thermal Energy Storage' in CSP plants.

Important point: Reflective surfaces in concentrating systems can cause accidental fires or blinding glares. All CSP sites must follow authorised optical safety and fire prevention protocols.

Module 4 — PV System Design and Integration

Explain Stand-alone System Design and state one correct method, application or precaution.−

Off-grid systems require batteries to store energy for use when the sun is not shining. Sizing involves estimating the daily energy demand, determining the required PV array capacity, and selecting a battery bank and charge controller to prevent overcharging.

Answer framework: Calculate the required 'Battery Capacity' (Ah) for a given load and autonomy period; explain the role of a 'Charge Controller'.

Important point: Lead-acid and Lithium batteries require proper ventilation and protection. Do not install battery banks in enclosed spaces without authorised gas venting and fuse protection.

Explain Grid Integration and Safety and state one correct method, application or precaution.−

Grid-connected systems use net metering to export excess power to the utility. The 'Balance of System' (BOS) includes mounting structures, cables, and protection devices. Safety is paramount during installation, requiring proper grounding and lightning protection.

Answer framework: Explain the concept of 'Net Metering'; list the main 'BOS' components of a grid-connected PV system.

Important point: Grid-tied systems must automatically disconnect during a power outage (anti-islanding). Do not connect to the grid without authorised utility approval and certified grid-tie equipment.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Solar Radiation and Measurement.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Solar Photovoltaic (PV) Technology.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Solar Thermal Systems.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: PV System Design and Integration.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Solar constant; Solar radiation: Extraterrestrial and Terrestrial; Solar geometry: Zenith, Azimuth, Tilt angles; Measurement: Pyranometer, Pyrheliometer; Estimation of solar radiation on tilted surfaces..
2. Develop a complete answer or practical plan for: Solar cell physics: P-N junction, I-V characteristics, Fill factor, Efficiency; PV module, panel and array; Cell types: Monocrystalline, Polycrystalline, Thin film; MPPT and Inverters..
3. Develop a complete answer or practical plan for: Principles of solar thermal conversion; Flat plate collectors (FPC); Concentrating collectors: Parabolic trough, Dish, Tower; Solar water heating; Solar thermal power plants (CSP)..
4. Develop a complete answer or practical plan for: Stand-alone PV system design: Load estimation, Battery sizing, Charge controller; Grid-connected systems: Net metering, Grid-tie inverters; Balance of System (BOS); Installation and safety..

LAST-MILE REVISION

Quick Revision

Module 1: Solar Radiation and Measurement

Solar constant; Solar radiation: Extraterrestrial and Terrestrial; Solar geometry: Zenith, Azimuth, Tilt angles; Measurement: Pyranometer, Pyrheliometer; Estimation of solar radiation on tilted surfaces.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Solar Photovoltaic (PV) Technology

Solar cell physics: P-N junction, I-V characteristics, Fill factor, Efficiency; PV module, panel and array; Cell types: Monocrystalline, Polycrystalline, Thin film; MPPT and Inverters.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Solar Thermal Systems

Principles of solar thermal conversion; Flat plate collectors (FPC); Concentrating collectors: Parabolic trough, Dish, Tower; Solar water heating; Solar thermal power plants (CSP).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: PV System Design and Integration

Stand-alone PV system design: Load estimation, Battery sizing, Charge controller; Grid-connected systems: Net metering, Grid-tie inverters; Balance of System (BOS); Installation and safety.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Solar Radiation Fundamentals	The Sun is a continuous source of energy.
Measurement and Estimation	Solar radiation is measured using a Pyranometer (for global radiation) and a Pyrheliometer (for beam radiation).
Solar Cell Operation	A solar cell is a P-N junction that converts photons into electron-hole pairs.
PV Modules and Electronics	Cells are connected in series/parallel to form modules and arrays.
Solar Collectors	Solar thermal systems collect heat from the sun.
Solar Thermal Applications	Common applications include solar water heaters, solar cookers, and solar dryers.
Stand-alone System Design	Off-grid systems require batteries to store energy for use when the sun is not shining.
Grid Integration and Safety	Grid-connected systems use net metering to export excess power to the utility.

LEARNING RESOURCES

References

Recommended solar power references

1. S. P. Sukhatme and J. K. Nayak, Solar Energy: Principles of Thermal Collection and Storage.
2. G. D. Rai, Solar Energy Utilization.
3. C. S. Solanki, Solar Photovoltaics: Fundamentals, Technologies and Applications.
4. H. P. Garg and J. Prakash, Solar Energy: Fundamentals and Applications.

Approved public solar power sources

- https://nptel.ac.in/courses/115103123
- https://onlinecourses.nptel.ac.in/noc20_ph14/preview

These sources provide the verified study basis while official Course 6034A detail is unavailable; they do not replace official plant designs, authorised installation standards, certified equipment specifications or authorised utility plans.